Dummit & Foote Notes

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Contents

0 Preliminaries

0.1 Basics

◆ Order/Cardinality: denoted as $|A|$ for a set A. If A is finite, then $|A|$ is the number of elements in A.
◆ Cartesian Product: The Cartesian product of two sets A and B is the collection of elements

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

◆ Function: Denoted as $f : A \to B$ for sets A and B. The value of f at a is denoted as $f(a)$ for all $a \in A$. Moreover, to denote $f(a) = b$, we write $f : a \mapsto b$, or $a \mapsto b$.
◆ Domain: The set A in the function $f : A \to B$.
◆ Codomain: The set B in the function $f : A \to B$.
◆ Range/Image: The subset of B defined as

$$f(A) = \{b \in B \mid f(a) = b \text{ for some } a \in A\}.$$

◆ Preimage/Inverse Image: For $C \subseteq B$, the subset of A given by

$$f^{-1}(C) = \{a \in A \mid f(a) \in C\}.$$

◆ Fiber: For $b \in B$, the preimage of b, i.e., $f^{-1}(b)$.

◆ Identity Map: The function on a set A defined as $\mathrm{id}_A : A \to A$ where $\mathrm{id}_A(a) = a$ for all $a \in A$.
◆ Composition Map: Given functions $f : A \to B$ and $g : B \to C$, it is the map $g \circ f : A \to C$ is defined as

$$(g \circ f)(a) = g(f(a)) \quad \text{for all } a \in A.$$

Let $f : A \to B$ be a function.
◆ Injective/Injection: for all $a_1, a_2 \in A$, $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$.
◆ Surjective/Surjection: for all $b \in B$, there exists $a \in A$ such that $f(a) = b$, i.e., $f(A) = B$.
◆ Bijective/Bijection: f is both injective and surjective.
◆ Left Inverse: A function $g : B \to A$ is a left inverse of $f : A \to B$ if $g \circ f = \mathrm{id}_A$.
◆ Right Inverse: A function $h : B \to A$ is a right inverse of $f : A \to B$ if $f \circ h = \mathrm{id}_B$.

♣ Proposition 0.1 ➤ Characterizations of Injectivity, Surjectivity, and Bijectivity

Let $f : A \to B$ be a function.
(1) f is injective if and only if f has a left inverse.
(2) f is surjective if and only if f has a right inverse.
(3) f is a bijection if and only if there exists $g : B \to A$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.
(4) If A and B are finite sets with $|A| = |B|$, then the following are equivalent:
  (a) f is bijective.
  (b) f is injective.
  (c) f is surjective.

Proof.
(1) ➡ Suppose f has a left inverse $g : B \to A$. Then for all $a_1, a_2 \in A$, if $f(a_1) = f(a_2)$, then

$$a_1 = g(f(a_1)) = g(f(a_2)) = a_2,$$

so f is injective.
⬅ Suppose f is injective. Then for each $b \in f(A)$, there exists a unique $a \in A$ such that $f(a) = b$. Now, fix some $a_0 \in A$. Then we can define a function $g : B \to A$ as

$$g(b) = \begin{cases} a & \text{if } b \in f(A) \text{ and } f(a) = b, \\ a_0 & \text{if } b \in B \setminus f(A). \end{cases}$$

To see g is well-defined, note that if $b \in f(A)$, then by injectivity of f, there is a unique $a \in A$ such that $f(a) = b$. Then for all $a \in A$,

$$g(f(a)) = a,$$

so $g \circ f = \mathrm{id}_A$ and g is a left inverse of f.

(2) ➡ Suppose f has a right inverse $h : B \to A$. Then for all $b \in B$, we may set $a = h(b) \in A$. Then $f(a) = f(h(b)) = b$, so f is surjective.

⬅ Suppose f is surjective. Then $f^{-1}(b)$ is nonempty for all $b \in B$. Define a function $h : B \to A$ by choosing some $a \in f^{-1}(b)$ for each $b \in B$, where this choice is possible by the axiom of choice. Then for all $b \in B$,

$$f(h(b)) = b,$$

so $f \circ h = \mathrm{id}_B$ and h is a right inverse of f.

(3) ➡ Suppose f is bijective. Then by (1) and (2), f has a left inverse $g : B \to A$ and a right inverse $h : B \to A$. Then for all $b \in B$,

$$g(b) = g(f(h(b))) = h(b),$$

where $g(f(h(b))) = g(b)$ because $f(h(b)) = b$ from h being a right inverse of f, and $g(f(h(b))) = h(b)$ because g is a left inverse of f. Thus, $g = h$ and $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.

⬅ Suppose there exists $g : B \to A$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$. Then by (1) and (2), f is injective and surjective, so f is bijective.

(4)
- If f is bijective, then by (1) and (2), f is injective and surjective.
- If f is injective, then distinct elements of A map to distinct elements of B. Since $|A| = |B|$, this implies that f is surjective, so f is bijective.
- If f is surjective, then all elements of B are mapped to by elements of A. Since $|A| = |B|$, this implies that f is injective, so f is bijective. □

◆ Inverse: The function $g : B \to A$ in Proposition 0.1 (3), denoted as $f^{-1} : B \to A$.

◆ Permutation: A bijection $f : A \to A$.

◆ Restriction: Let $A \subseteq B$ and $f : B \to C$ be a function. The restriction of f to A is the function $f|_A : A \to C$ defined as

$$f|_A(a) = f(a) \quad \text{for all } a \in A.$$

◆ Extension: Let $A \subseteq B$ and $g : A \to C$ be a function. The extension of g to B is a function $f : B \to C$ such that $f|_A = g$. This function need not be unique, and may not even exist.

Let A be a set.

◆ Binary Relation: A subset $R \subseteq A \times A$, where we write $a \sim b$ if $(a, b) \in R$.

◆ Reflexive: A relation R is reflexive if $a \sim a$ for all $a \in A$.

◆ Symmetric: A relation R is symmetric if $a \sim b$ implies $b \sim a$ for all $a, b \in A$.

◆ Transitive: A relation R is transitive if $a \sim b$ and $b \sim c$ implies $a \sim c$ for all $a, b, c \in A$.

◆ Equivalence Relation: A relation R that is reflexive, symmetric, and transitive.

◆ Equivalence Class: For $a \in A$, the equivalence class of a is the set

$$[a] = \{b \in A \mid b \sim a\}.$$

Elements of the $[a]$ are said to be equivalent to a and are representatives of $[a]$.

◆ Partition: Let I be an indexing set. A partition of A is a collection of nonempty subsets $\{A_i\}_{i \in I}$ such that

(1) $A = \bigcup_{i \in I} A_i$.

(2) $A_i \cap A_j = \varnothing$ for all $i, j \in I$ with $i \neq j$.

♣ Proposition 0.2 ➤ Fundamental Theorem of Equivalence Relations

Let A be a nonempty set.

(1) If $\sim$ is an equivalence relation on A, then the set of equivalence classes of $\sim$ form a partition of A.

(2) Let I be an indexing set. If $\{A_i \mid i \in I\}$ is a partition of A, then there is an equivalence relation on A whose equivalence classes are the sets A_i for $i \in I$.

Proof.

(1) Let $\sim$ be an equivalence relation on A, and put

$$B = \bigcup_{a \in A} [a]$$

to be the union of all equivalence classes of $\sim$. For any $a \in A$, we have $a \sim a$ by reflexivity, so $a \in [a]$ and thus $a \in B$. Hence, $A \subseteq B$. Conversely, let $b \in B$. Then $b \in [a]$ for some $a \in A$, so $b \sim a$ and thus $b \in A$. Hence, $B \subseteq A$ and thus $A = B$.

Now, let $a, b \in A$. If $[a] \cap [b] = \varnothing$, then we are done. Otherwise, let $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$, so by symmetry and transitivity, we have $a \sim c$ and $c \sim b$, which implies $a \sim b$. Now, let $x \in [a]$. Then $x \sim a$ and $a \sim b$, so by transitivity, we have $x \sim b$ and thus $x \in [b]$. Hence, $[a] \subseteq [b]$. A similar argument shows that $[b] \subseteq [a]$, so $[a] = [b]$, and thus the equivalence classes of $\sim$ form a partition of A.

(2) Let $\{A_i \mid i \in I\}$ be a partition of A. Define a relation $\sim$ on A by $a \sim b$ if and only if there exists $i \in I$ such that $a, b \in A_i$. Then $\sim$ is reflexive because for any $a \in A$, there exists $i \in I$ such that $a \in A_i$, so $a \sim a$. Moreover, $\sim$ is symmetric because if $a \sim b$, then there exists $i \in I$ such that $a, b \in A_i$, so $b \sim a$. Finally, $\sim$ is transitive because if $a \sim b$ and $b \sim c$, then there exist $i, j \in I$ such that $a, b \in A_i$ and $b, c \in A_j$. Since $b \in A_i \cap A_j$, we must have $i = j$ because the A_i are disjoint for $i \neq j$. Hence, $a, c \in A_i$, so $a \sim c$. Therefore, $\sim$ is an equivalence relation, and its equivalence classes are precisely the sets A_i for $i \in I$. $\square$